

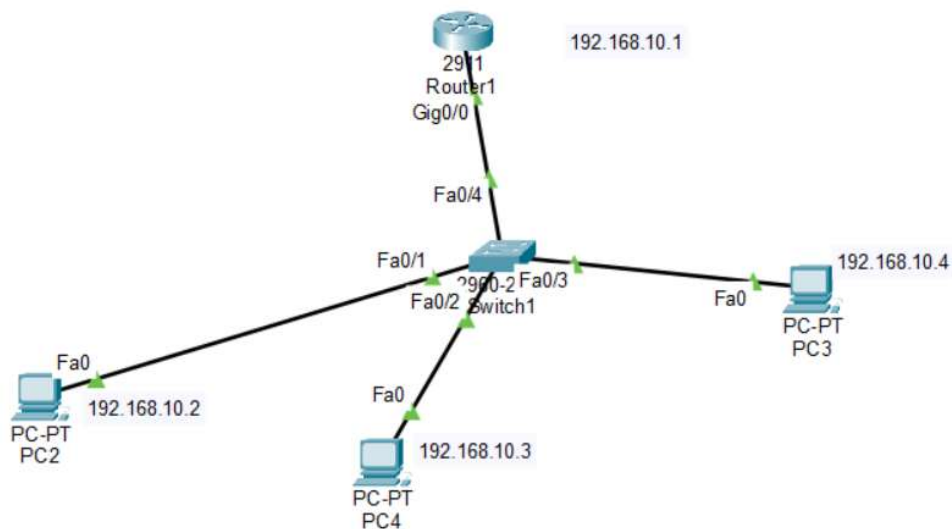
Common Network Issues theory

Q1. List 5 common hardware issues that can cause network problems in a typical office setup, and for each, describe one symptom that helps identify it.

No.	Hardware issue	Symptom that helps identify it
1	Faulty or loose Ethernet cable	Link light on the PC's NIC or switch port is off, or the connection drops when the cable is moved.
2	Failed switch port	One device has no link light, while other devices on the same switch work fine.
3	Router or switch power failure / overheating	All lights are off, or the device reboots randomly and the whole office loses connectivity.
4	Faulty NIC (network card)	PC shows "network cable unplugged" or no adapter, even with a known-good cable.
5	Damaged RJ-45 connector or poor crimping	Connection is sometimes, only transfers 10/100 Mbps instead of 1 Gbps, and shows packet loss.

Q2. Open Cisco Packet Tracer or any network simulator, set up a small network with 3 PCs, 1 switch, and 1 router, and intentionally disconnect one device. Use the simulator's troubleshooting tools to identify which device is faulty and document the steps you took.

Network setup



Connection between PCs checked using ping command, ping successful.

```
PC2
Physical | Config | Desktop | Programming | Attributes |
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=10ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fault introduced

The cable between PC2 and the switch was disconnected.

Troubleshooting steps

1. **Checking the topology:** the link between PC2 and the switch shows a red link light instead of green.
2. **Ping from PC2:** ping 192.168.1.1 returns "Request timed out".
3. **Ping from a working PC:** PC0 can reach the router and PC1, but PC0 cannot reach PC2. This isolates the problem to PC2.

```
C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

4. **Check PC2's IP configuration** with ipconfig: the IP address and gateway are correct, so it is not a configuration issue.

```

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:D3FF:FE58:DD68
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.10.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

```

5. **Check the switch:** show ip interface brief shows the port connected to PC2 as down/down.

```

Switch#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status
FastEthernet0/1	unassigned	YES	manual	down
FastEthernet0/2	unassigned	YES	manual	up
FastEthernet0/3	unassigned	YES	manual	up
FastEthernet0/4	unassigned	YES	manual	up
FastEthernet0/5	unassigned	YES	manual	down
FastEthernet0/6	unassigned	YES	manual	down
FastEthernet0/7	unassigned	YES	manual	down
FastEthernet0/8	unassigned	YES	manual	down

6. **Conclusion:** the fault is the PC2 link (cable or port). The router, switch and other PCs are working.
7. **Fix:** reconnect the cable and ping again to verify that PC2 can reach the router and other PCs.

Q3. Create a troubleshooting checklist for diagnosing slow internet issues in a home WiFi network, including at least 8 steps covering both hardware and software checks.

1. **Check the router lights:** power, internet/WAN and WiFi lights should be normal, with no red or warning indicators.
2. **Check cables:** make sure the modem-to-router cable and power cables are firmly connected and undamaged.
3. **Restart the modem and router:** power them off for about 30 seconds, then turn them back on.
4. **Test on another device:** see whether the slowness affects all devices or only one.
5. **Compare WiFi and wired:** connect a laptop by Ethernet. If it is fast, the problem is WiFi signal or interference.
6. **Check signal strength and distance:** move closer to the router and look for walls or devices causing interference.
7. **Check connected devices:** disconnect unknown devices and pause heavy downloads or streaming.
8. **Run a speed test** (for example speedtest.net) and compare the result with the speed of your internet plan.
9. **Run ping tests:** ping 8.8.8.8 checks connectivity and ping google.com checks DNS. High latency or packet loss indicates a network problem.
10. **Check device settings:** look for a VPN, proxy or background updates, and update the network driver.

11. **Change the WiFi channel or band:** try 5 GHz instead of 2.4 GHz to avoid congestion.
12. **Update the router firmware** to the latest version.
13. **Contact the ISP** if the wired connection is also slow.

Q4. You receive a complaint that a user cannot access Instagram and YouTube on their laptop, but WhatsApp Web works fine. Write down the first 5 troubleshooting steps you would perform to identify if the issue is with hardware, software, or network configuration.

1. **Confirm the scope:** try the same sites on another device on the same network, and on mobile data. This shows whether the problem is the laptop or the network.
2. **Ping by IP and by name:** run `ping 8.8.8.8`, then `ping www.youtube.com`. If the IP works but the name fails, it is a DNS problem.
3. **Test DNS resolution:** run `nslookup instagram.com` and `nslookup youtube.com` and check whether they resolve to a correct IP address.
4. **Check the laptop settings:** review the hosts file, proxy settings, VPN, browser extensions and DNS server settings for anything blocking those sites.
5. **Check network-level blocking:** look at router or firewall rules, parental controls and content filters. Try changing the DNS to 8.8.8.8 to see if it fixes the problem.

Conclusion

Because WhatsApp Web works, the physical connection and internet access are fine, so this is not a hardware problem. When only specific sites fail, the usual causes are DNS, a blocking rule (firewall or filter), or a laptop setting such as a proxy or hosts file entry.